CHAPTER 14

Derivation of State Graphs and Tables

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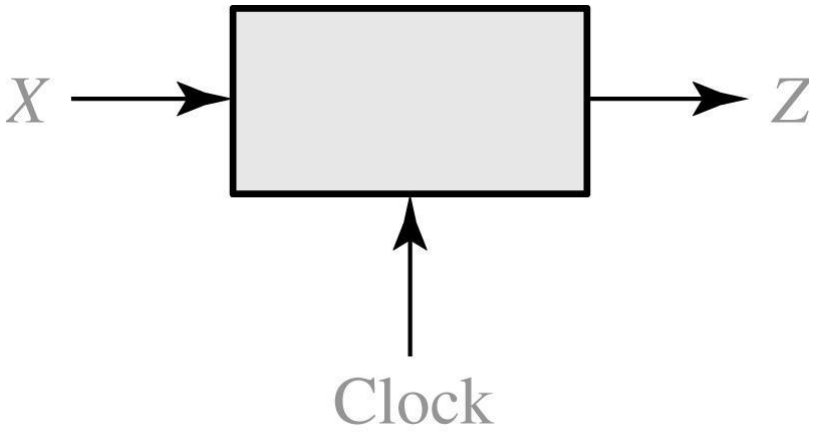
Objective

- Given a problem, how to derive the state graph and table
  - Define the specific states for the problem
- Design Moore machine and Mealy machine for the same problem

Design of Sequence Detector - Mealy Machine (1)

Ex: Fig 14.1 Sequence Detector to be Designed



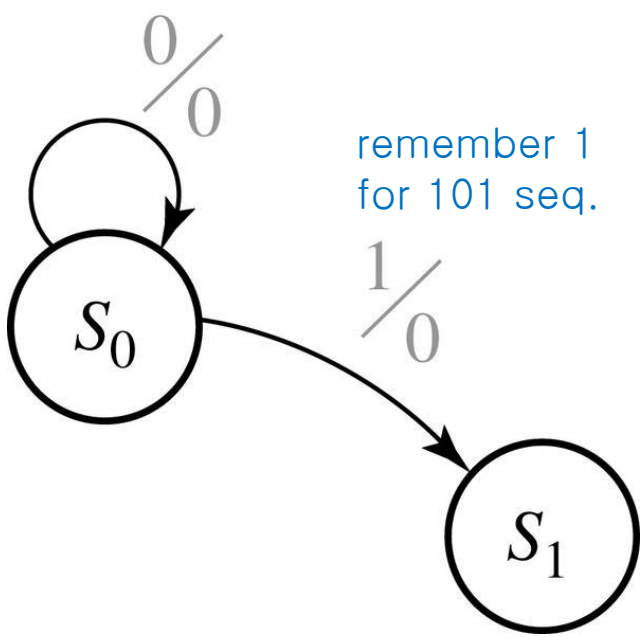
Ex: Z=1, if input sequence ends in 101.

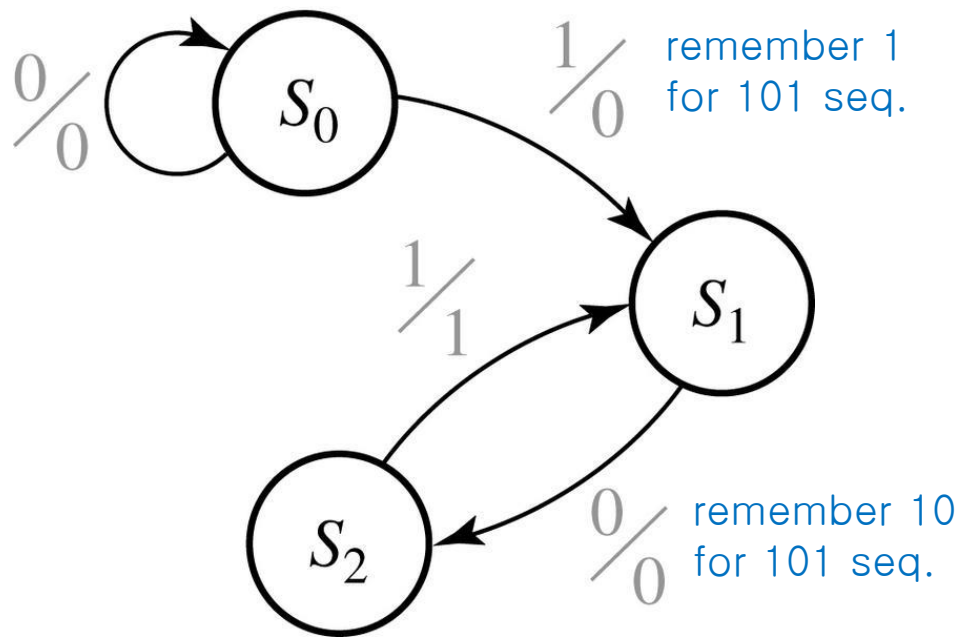
X =	0	0	1	1	0	1	1	0	0	1	0	1	0	1	0	0
Z =	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0
(time:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15)

Design of Sequence Detector - Mealy Machine (2)

Fig 14.2 and 14.3 : **Formation of State Graph**

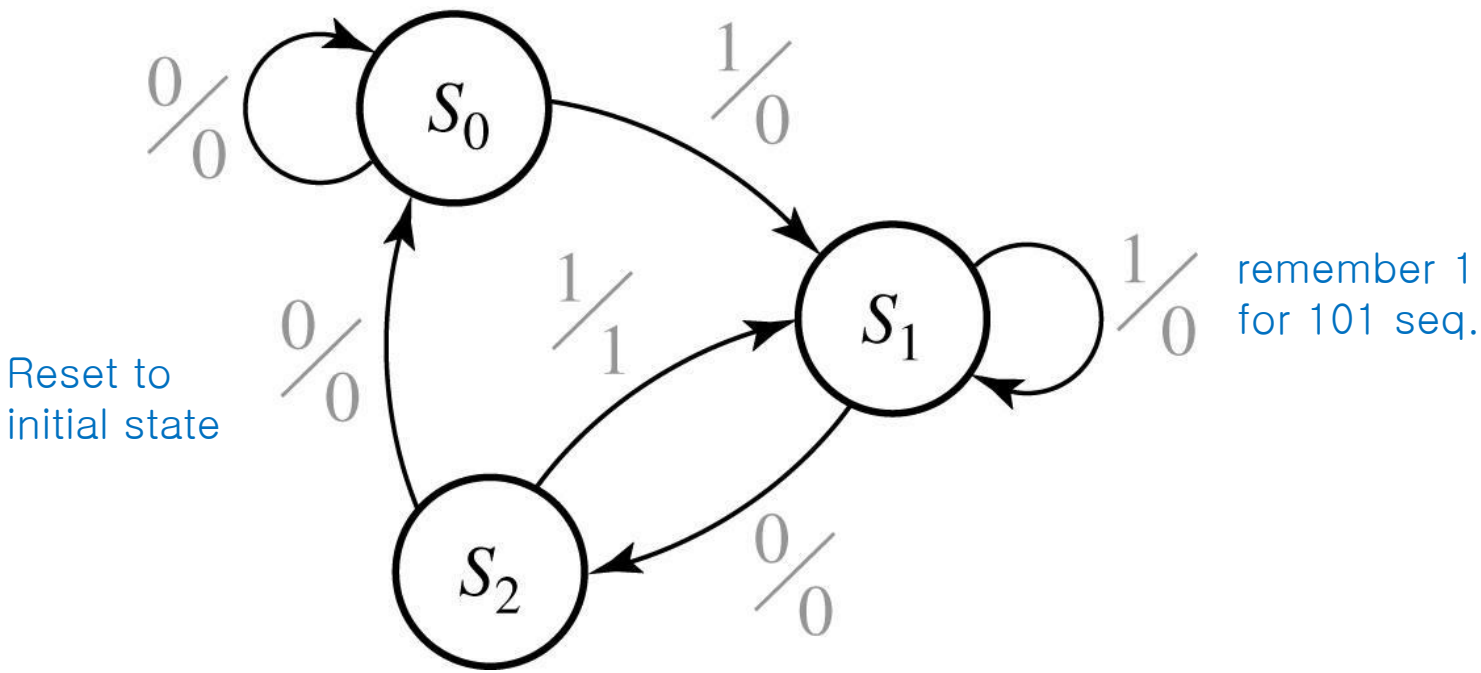
- set states, s_0, s_1, s_2…
- starting an initial state s_0, form the graph w.r.t inputs





Design of Sequence Detector - Mealy Machine (3)

Fig 14.4 Mealy State Graph for Sequence Detector



Design of Sequence Detector - Mealy Machine (4)

Table 14-1, State Table

Present state	Next State		Present Output	
	X = 0	X = 1	X = 0	X =1
S_0	S_0	S_1	0	0
S_1	S_2	S_1	0	0
S_2	S_0	S_1	0	1

Table 14-2, Transition Table with State Assignment

AB	$A^+ B^+$		Z	
	X = 0	X = 1	X = 0	X =1
00	00	01	0	0
01	10	01	0	0
10	00	01	0	1

Design of Sequence Detector - Mealy Machine (5)

Next states Maps, and map for for the output function Z

AB \ X	0	1
00	0	0
01	1	0
11	X	X
10	0	0

$A^+ = X'B$

AB \ X	0	1
00	0	1
01	0	1
11	X	X
10	0	1

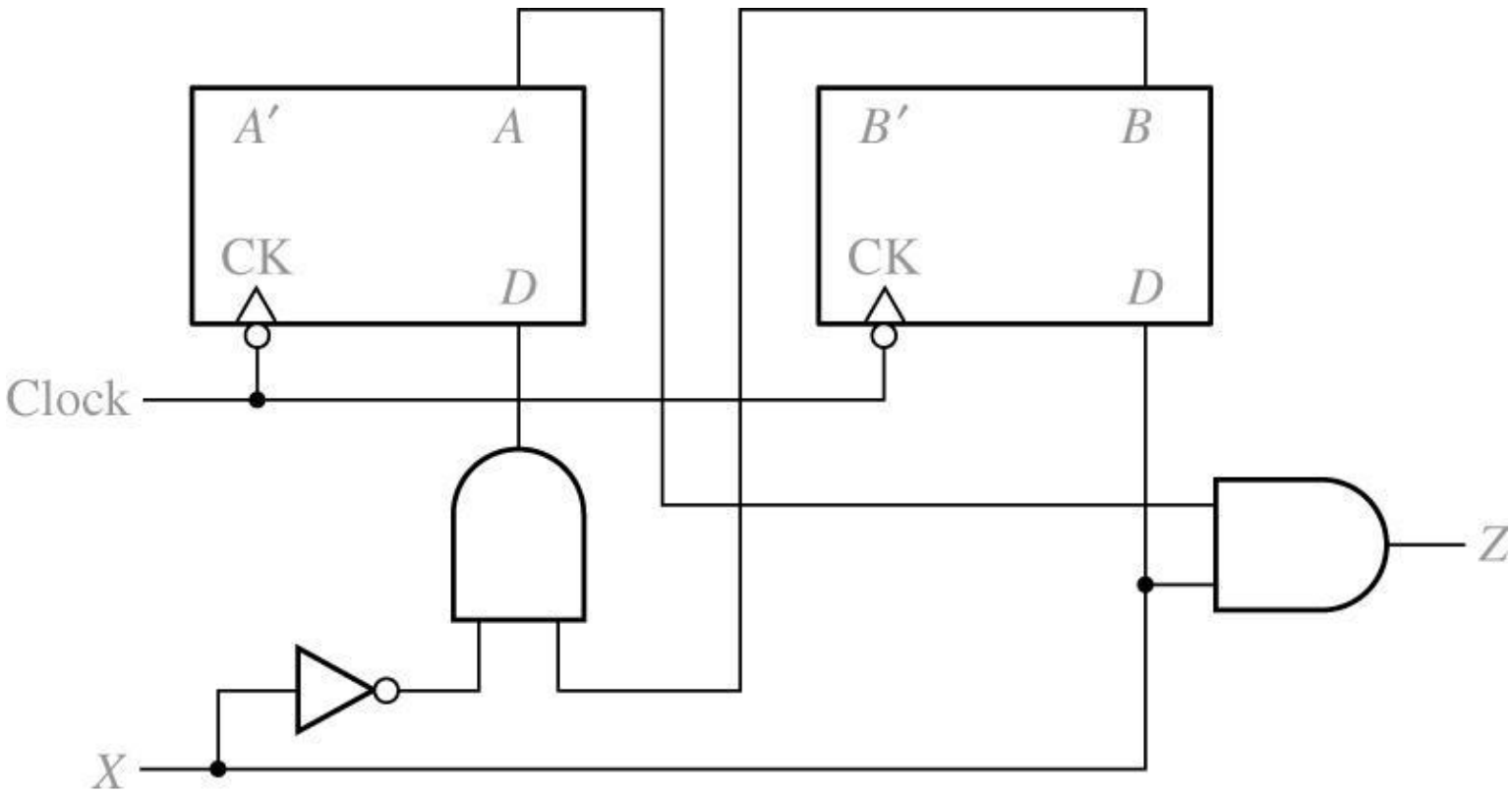
$B^+ = X$

AB \ X	0	1
00	0	0
01	0	0
11	X	X
10	0	1

$Z = XA$

Design of Sequence Detector - Mealy Machine (6)

Fig 14.5: Final Circuit using D Flip-Flop

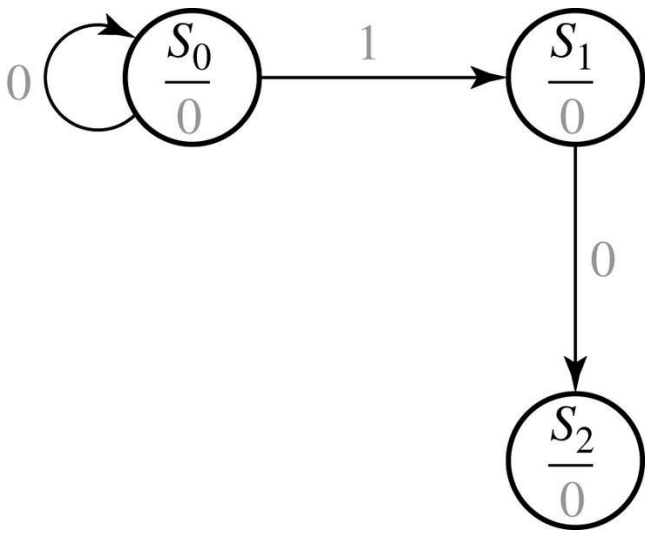


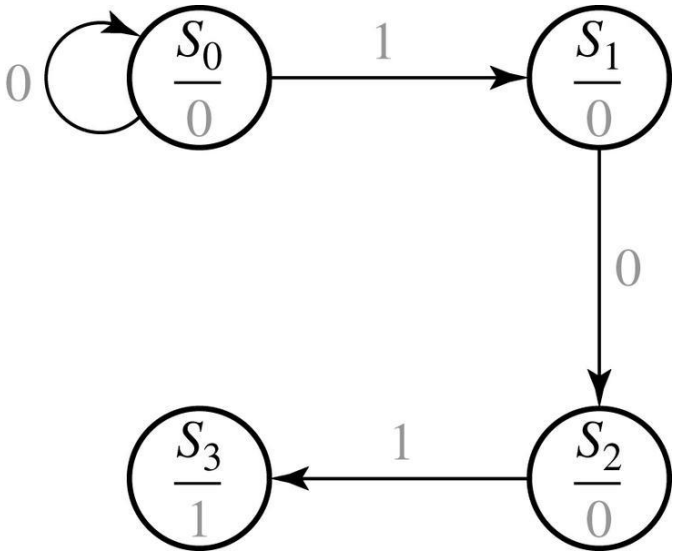
Design of Sequence Detector - Moor Machine (1)

Moore Machine Design Process

- the output is written with the state instead of with the transition

- Since Moore machine has an output in each state, we have to make more states, but Mealy machine can have states with two outputs because the output is designated on the transition.

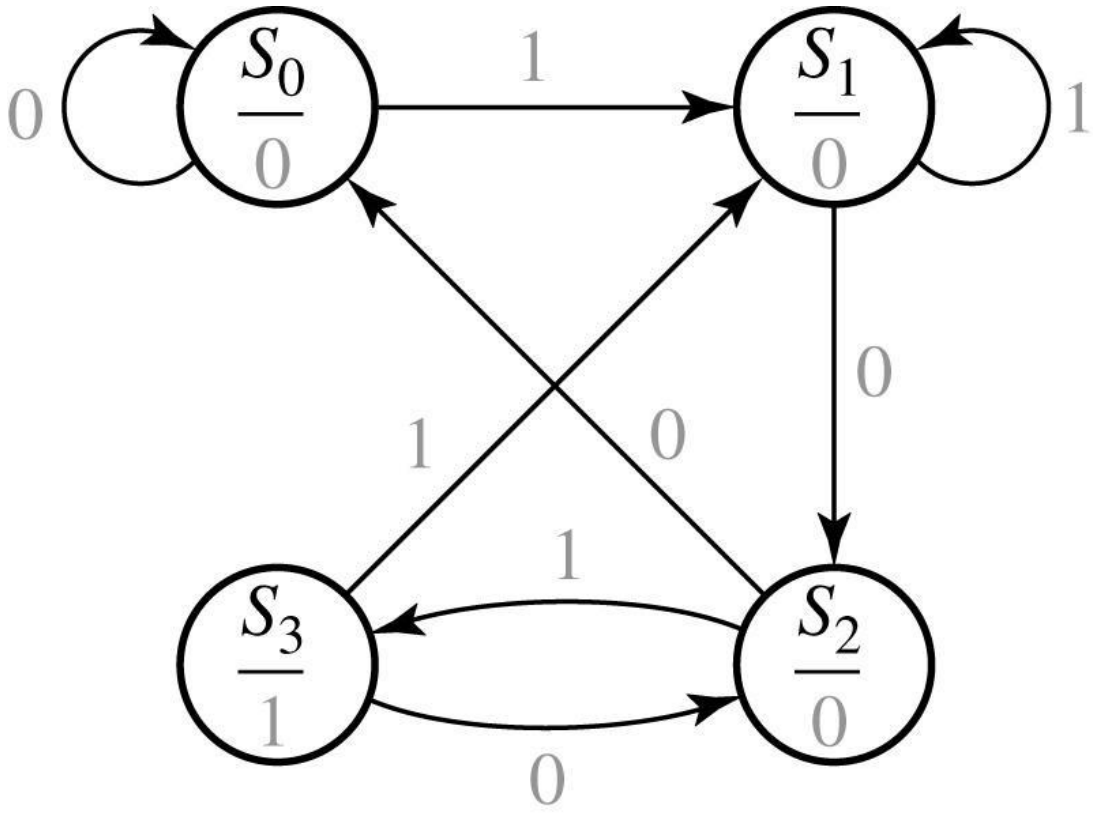




Create new state with output 1

Design of Sequence Detector - Moor Machine (2)

Fig 14.6 Moore State Graph for Sequence Detector



Design of Sequence Detector - Moor Machine (3)

Table 14-3 State Table

Present state	Next State X = 0	Next State X = 1	Present Output (Z)
S_0	S_0	S_1	0
S_1	S_2	S_1	0
S_2	S_0	S_3	0
S_3	S_2	S_1	1

Table 14-4 Transition Table with State assignment

AB	$A^+ B^+$ X = 0	$A^+ B^+$ X = 1	Z
00	00	01	0
01	11	01	0
11	00	10	0
10	11	01	1

The circuit design using FF's is similar to that of Mealy machine.

- next state maps and output map

14.2 More Complex Design Problems

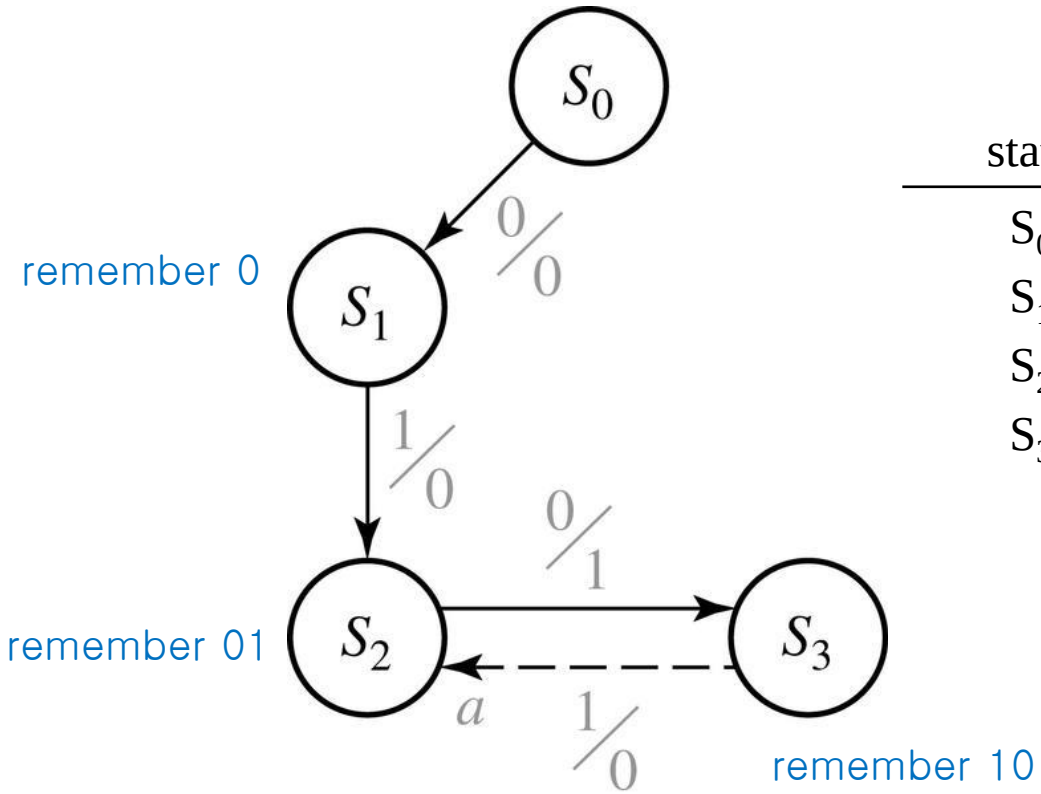
The circuit to be designed (Mealy)

Output Z=1 if input sequence ends in either 010 or 1001

X=	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0
				↑		↑		↑	↑				↑		↑		
				a		b		c	d				e		f		
Z=	0	0	0	1	0	1	0	1	1	0	0	0	1	0	1	0	0

14.2 More Complex Design Problems

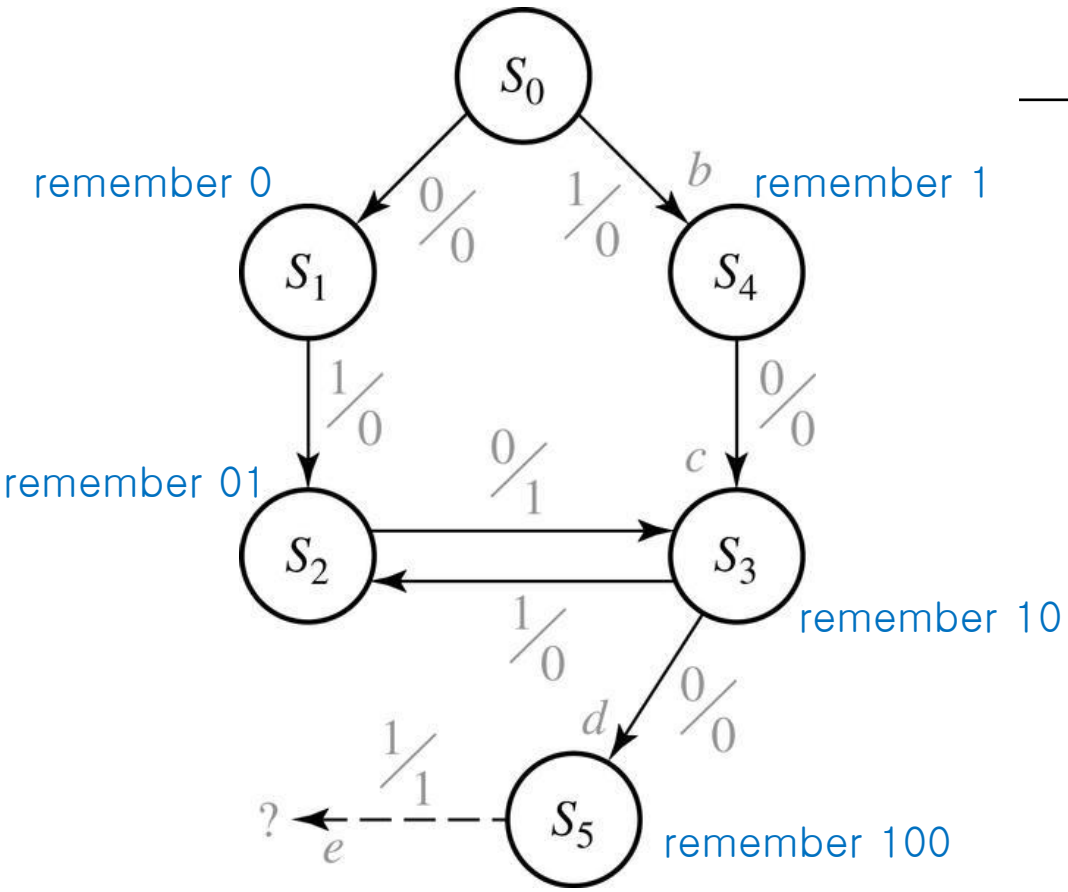
Fig 14.7 formation of state graph (step1)



state	sequence received
S_0	Reset (no input)
S_1	0
S_2	01
S_3	010

14.2 More Complex Design Problems

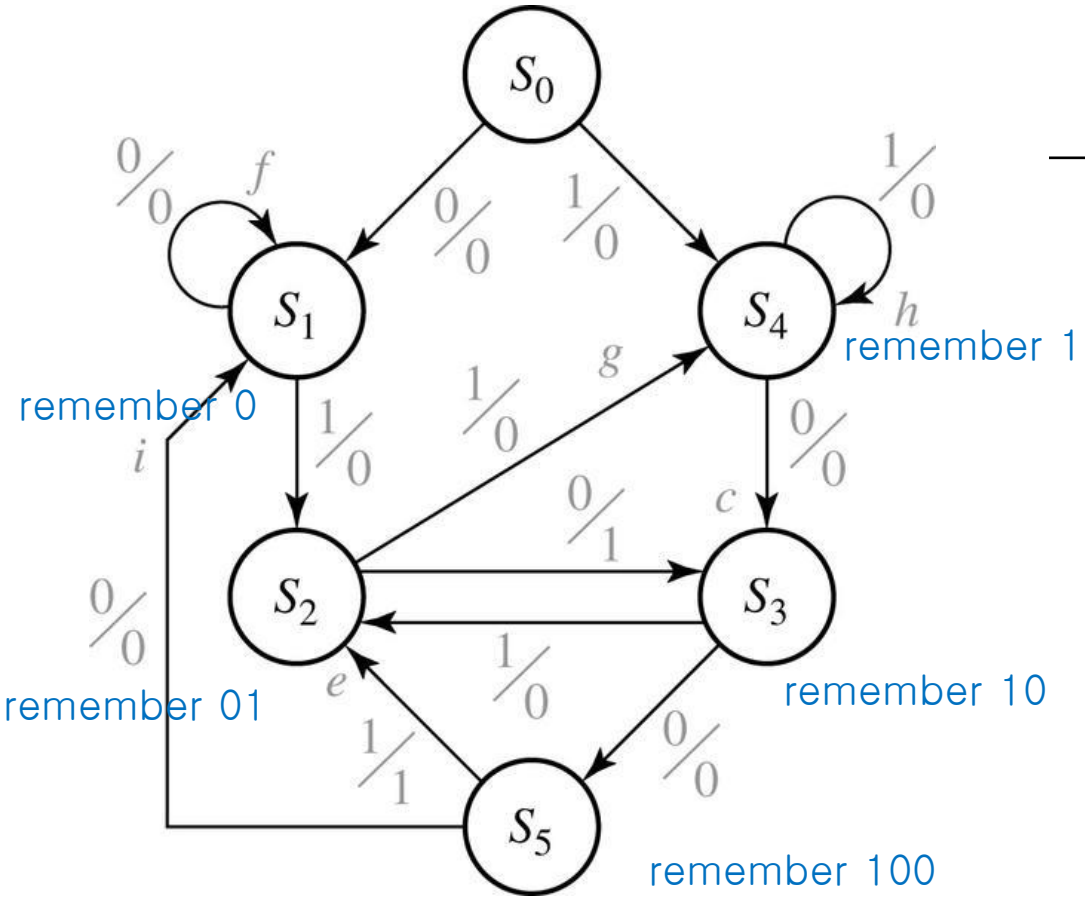
Fig 14.8 formation of state graph (step2)



state	sequence ends in
S_0	reset
S_1	0 (but not 10)
S_2	01
S_3	10
S_4	1 (but not 01)
S_5	100

14.2 More Complex Design Problems

Fig 14.9 Completed State Graph for a Sequence Detector to be Designed



state	sequence ends in
S_0	reset
S_1	0 (but not 10)
S_2	01
S_3	10
S_4	1 (but not 01)
S_5	100

Guidelines for Construction of State Graphs

1. **Construct some sample input and output sequences** to make sure that you understand the problem statement.

2. **Determine under what conditions the circuit should reset to its initial state.**

3. If only one or two sequences lead to a non-zero output, a **good start is to construct a partial state graph** for those sequences.

4. **Determine what sequences or groups of sequences must be remembered by the circuit** and set up states accordingly.

5. Each time you add an arrow to the state graph, **check it can go to one of the previously defined states or whether a new state must be added**.

6. Check your state graph to **make sure there is one and only one path leaving each state** for each combination of values of the input variables.

7. When your state graph is complete, test it by applying the input sequences and making sure the output sequences are correct.